



JSP III

1. Std actions
2. Bean related std actions
3. MVC architecture

- JSP standard actions are special elements.
- These tags use XML like syntax and are processed by the JSP container during page translation.
- Std actions are the tags starting with jsp prefix.
- Examples of std actions
 - Flow Control: actions like `jsp:forward` and `jsp:include` help in managing navigation between JSP pages
 - Java bean manipulation: Actions like `jsp:useBean`, `jsp:setProperty`, and `jsp:getProperty`
 - Dynamic Content Generation: Tags like `jsp:body`, `jsp:attribute`, and `jsp:text` enable you to dynamically define content based on various factors
 - Other actions: Additional tags like `jsp:plugin` and `jsp:param` etc

- JSP provides mechanisms to interact with beans created and access and update it's properties.
- Java beans.
 - java objects that have private properties which can be accessed using getter methods and updated using setter methods.
 - It should have default constructor compulsorily.
- Std actions
 - `<jsp:useBean id="userinfo" class="beans.User" scope="session" />` : Tries to locate the bean userinfo at session scope, if not found creates it
 - `<jsp:setProperty name="userinfo" property="uid" param="uid" />`: Reads value from request parameter and assigns to bean property
 - `<jsp:getProperty name="userinfo" property="fname" />`: Access fname property of existing bean.

- MVC (Model-View-Controller) is a widely used architectural design pattern for developing web applications.
- Model:
 - Represents the business data
 - Typically implemented as Java classes
- View:
 - Responsible for presenting information to the user.
 - Views are often implemented using JSP
 - Display data retrieved from the model
- Controller:
 - Handles incoming user requests.
 - Prepares data for the view and selects the appropriate view template to display.

- Benefits of MVC:
 - Separation of Concerns: Isolates data, business logic, and presentation into distinct components.
 - Flexibility: Changes to one component (e.g., the view) do not affect other components
 - Testability: Each layer (model, view, controller) can be tested independently, making development and debugging easier.
- Steps for MVC implementation